

KEYS DECIDE, VALUES DESCRIBE: ACCEPTANCE-AWARE KV CACHE COMPRESSION FOR SPECULATIVE DECODING

Anonymous authors

Paper under double-blind review

ABSTRACT

Speculative decoding accelerates language-model inference by asking a small draft model to propose tokens for verification by a larger target model, but standard implementations retain a separate key-value (KV) cache for each model. We study how the draft cache can be compressed while preserving the quantity that directly controls speculative speed: token acceptance. Our central observation is that acceptance is strongly asymmetric in key and value precision. Across six target-draft pairs, two text distributions, and contexts from 1K to 16K, 8-bit keys with 4-bit values (K8V4) outperform the equal-memory K4V8 assignment in 19 of 20 matched comparisons. On Qwen2.5-3B/1.5B, K8V4 improves acceptance by 0.312 at 1K and 0.336 at 4K while reducing total target-plus-draft KV memory by 27.0%. Controlled Gaussian perturbations expose the same mechanism: key noise changes attention routing and rapidly destroys top-1 stability, whereas value noise is tolerated at much larger magnitudes. Based on this observation, SPECKV profiles each draft layer and cache component under the actual speculative loop, then selects mixed precision under an acceptance-drop budget. The target verifier is never quantized; we separately audit finite-precision differences between batched verification and tokenwise BF16 decoding.

1 INTRODUCTION

Speculative decoding converts several serial target-model decoding steps into a single parallel verification step by sampling proposals from a cheaper draft model (Leviathan et al., 2023; Chen et al., 2023). The method is distribution preserving, but its systems footprint includes two model states and two growing KV caches. This duplication becomes consequential at long context, where each generated token reads the prefix cache and cache capacity limits the number of concurrently served requests.

Quantizing the draft cache seems straightforward, but generic model-quality metrics do not reveal whether a compressed cache remains a useful speculator. A small distributional error can move the draft’s top proposal away from the target’s token and terminate an entire speculative block. Conversely, errors in directions irrelevant to the verifier may have little effect on acceptance. This motivates an acceptance-aware rather than reconstruction-aware objective.

Our investigation starts from the attention decomposition

$$A_\ell = \text{softmax}\left(\frac{Q_\ell K_\ell^\top}{\sqrt{d_h}} + M\right), \quad O_\ell = A_\ell V_\ell W_{O,\ell}. \quad (1)$$

Keys determine the routing distribution A_ℓ , whereas values determine the payload read after routing. We hypothesize that coarse key quantization is more likely to redirect attention and cascade through subsequent layers than an equal-memory reduction in value precision. This role asymmetry is compatible with prior KV quantization analyses (Liu et al., 2024; Hariri et al., 2025), but we ask a different operational question: *which errors can a draft model tolerate without losing speculative acceptance?*

Our findings support three claims. First, Gaussian key perturbations are dramatically more destructive to next-token stability than value perturbations. Second, K8V4 dominates K4V8 in 19 of 20 equal-

memory comparisons across six model pairs; the only reversal is an OLMo difference of -0.002 whose confidence interval includes zero. Third, a conservative layer-wise allocator retains 94.9–102.8% of native acceptance while saving 2.9–9.9% of total KV memory.

2 BACKGROUND AND PROBLEM FORMULATION

Let $p(\cdot \mid x_{\leq t})$ be the target distribution and $q(\cdot \mid x_{\leq t}; \hat{\mathcal{C}}_d)$ the draft distribution produced from a compressed draft cache $\hat{\mathcal{C}}_d$. Standard speculative sampling accepts proposals using the target correction rule; greedy speculative decoding accepts the longest proposal prefix matching target argmax tokens. In both cases, target verification is distribution preserving in exact arithmetic: compression affects efficiency through acceptance, not the target correction rule. In finite-precision greedy experiments, batched and token-wise target evaluation can choose different tokens when the largest BF16 logits are tied. We therefore report both sequence agreement and the target top-1 margin at every disagreement, distinguishing numerical ties from larger kernel-path differences.

For target and draft cache sizes M_t and M_d , native speculative decoding uses $M_t + M_d$. If layer ℓ stores keys and values with precisions b_ℓ^K, b_ℓ^V , the compressed draft cost is approximately

$$\widehat{M}_d \propto T \sum_{\ell=1}^{L_d} H_\ell^{KV} d_{h,\ell} (b_\ell^K + b_\ell^V), \quad (2)$$

plus one scale per token and KV head in our per-vector quantizer. Total saving is $(M_d - \widehat{M}_d)/(M_t + M_d)$; it is necessarily smaller than the percentage saved from the draft cache alone.

3 METHOD

3.1 DRAFT-CACHE FAKE QUANTIZATION

For a key or value vector x , we use symmetric per-vector fake quantization,

$$s(x, b) = \frac{\max_i |x_i|}{2^{b-1} - 1}, \quad \hat{x} = s \text{ clip}(\text{round}(x/s), -2^{b-1} + 1, 2^{b-1} - 1). \quad (3)$$

Only the draft KV state is quantized between cached decode steps. The target model and verifier remain in BF16. This simulator measures the algorithmic memory–acceptance tradeoff without conflating it with a particular packed-cache kernel.

The evaluator performs one target and one draft prefill per prompt. Each draft round starts from the retained draft prefix cache, while the target verifies the proposal in one cached forward pass. Accepted states are committed and rejected suffix states are cropped in place; neither model recomputes the long prefix. Target-greedy references are generated once outside the timed configuration loop. This cache-resident execution is essential at long context and prevents prefix recomputation from dominating either memory or runtime measurements.

3.2 ACCEPTANCE-AWARE SENSITIVITY PROFILES

We first evaluate a full-precision draft baseline. Then, for each selected layer ℓ , component $c \in \{K, V\}$, and candidate bit width $b \in \{4, 8\}$, we quantize only (ℓ, c) while leaving all other draft-cache entries in BF16. The measured risk is

$$r_{\ell,c,b} = \max(0, \alpha_{\text{BF16}} - \alpha_{\ell,c,b}), \quad (4)$$

where α is the end-to-end speculative acceptance rate. We additionally log accepted tokens per verification round, top-1 match, acceptance mass, Jensen–Shannon divergence, greedy target agreement, and the target top-1 margin at the first disagreement.

3.3 BUDGETED MIXED-PRECISION ALLOCATION

Given one-at-a-time risks, our transparent initial allocator solves an additive surrogate:

$$\min_{\{b_\ell^c\}} \widehat{M}_d \quad \text{s.t.} \quad \sum_{\ell,c} r_{\ell,c,b_\ell^c} \leq \epsilon, \quad b_\ell^c \in \{4, 8, 16\}. \quad (5)$$

Table 1: Cross-family evaluation. Each pair is tested at 1K and 4K context, followed by top-eight-layer K/V profiling and held-out allocation evaluation.

Family	Target	Draft
Qwen2.5	3B, 7B	1.5B, 3B
Qwen3	8B	4B
Llama 3	Llama-3.1-8B	Llama-3.2-3B
OLMo 2	7B	1B
SmolLM2	1.7B	360M

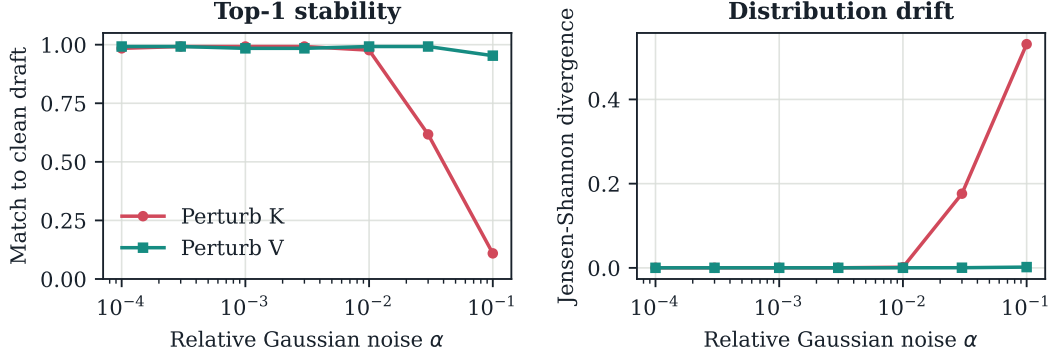


Figure 1: Controlled perturbations of the Qwen2.5-1.5B draft cache. Keys are far more sensitive than values under both top-1 stability and distribution drift.

The additive approximation is deliberately conservative and is validated by running the selected allocation through the complete speculative loop. A stronger version will model cross-layer interactions rather than assuming independent risks.

4 EXPERIMENTAL SETUP

We evaluate the six target–draft pairs in Table 1 using contiguous WikiText-2 validation blocks, four draft proposals per verification round, greedy target verification, and 16 generated tokens per prompt. Main 1K and 4K experiments use 32 and 12 prompts per pair. We add two shuffled C4 seeds for Qwen2.5-3B/1.5B, OLMo-2, and Llama, and exploratory Qwen2.5 runs at 8K and 16K. Sensitivity profiles use eight prompts per candidate; held-out allocation evaluation uses 32. All pairs were checked for tokenizer compatibility.

We report acceptance rate, accepted tokens per target verification, top-1 match, JS divergence, acceptance mass, draft-cache bytes, and total target-plus-draft KV bytes. Python fake quantization does not provide a valid kernel-level speed measurement, so we do not make throughput claims from this implementation.

5 RESULTS

5.1 KEYS ARE THE FRAGILE ROUTING STATE

Figure 1 perturbs all draft-cache keys or values with Gaussian noise scaled to the clean activation standard deviation. At $\alpha = 0.03$, key noise reduces top-1 agreement with the clean draft to 0.617 and raises JS divergence to 0.176. Value noise at the same magnitude retains 0.992 top-1 match with JS divergence 0.0004. At $\alpha = 0.1$, the corresponding top-1 matches are 0.109 for K and 0.953 for V. This large separation is consistent with routing errors in K changing A in Equation 1 before values are read.

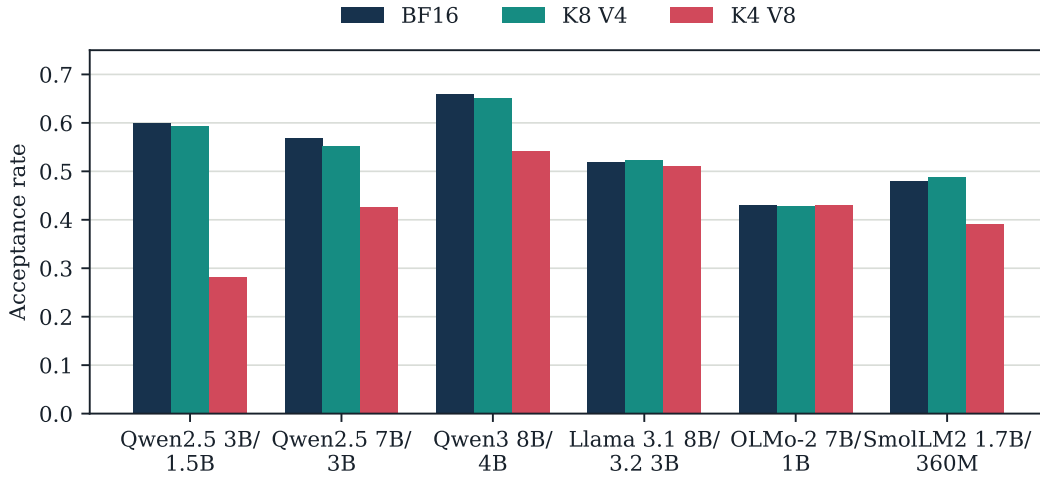


Figure 2: Acceptance at 1K WikiText context. Preserving key precision dominates preserving value precision at equal memory for five families and ties on OLMo.

Table 2: Equal-memory K8V4 versus K4V8 at 1K WikiText. Memory includes 16-bit per-vector scales.

Target / draft	K8 V4	K4 V8	Δ accept.	Total KV saved
Qwen2.5 3B/1.5B	0.594	0.282	+0.312	27.0%
Qwen2.5 7B/3B	0.551	0.426	+0.125	24.2%
Qwen3 8B/4B	0.651	0.542	+0.109	23.6%
Llama 3.1 8B/3.2 3B	0.523	0.512	+0.011	28.8%
OLMo-2 7B/1B	0.428	0.430	-0.002	12.3%
SmolLM2 1.7B/360M	0.487	0.390	+0.097	10.5%

5.2 EQUAL MEMORY DOES NOT IMPLY EQUAL ACCEPTANCE

Figure 2 isolates the core comparison at 1K context. K8V4 and K4V8 have identical average precision and save the same bytes, yet K8V4 wins in five of six families and ties OLMo within 0.002. Across all contexts and held-out distributions it wins 19 of 20 matched comparisons. The effect is largest for Qwen2.5-3B/1.5B: acceptance improves from 0.282 to 0.594 at 1K and from 0.300 to 0.636 at 4K. K8V4 saves 61.7% of the draft cache and 27.0% of combined target-plus-draft KV memory for this pair. Depending on model-size ratio, total savings range from 10.5% (SmolLM2) to 28.8% (Llama).

The asymmetry persists outside the main setting. On two shuffled C4 seeds, K8V4–K4V8 differences are +0.185, +0.282 for Qwen, +0.010, +0.074 for OLMo, and +0.026, +0.090 for Llama. For Qwen at 8K and 16K, the differences remain +0.260 and +0.140, respectively, although these exploratory long-context runs contain only four and two prompts.

5.3 A CONSERVATIVE LEARNED ALLOCATION

Profiling the top eight draft layers and selecting precision under a five-point additive acceptance budget gives deliberately conservative allocations. Across the six pairs, held-out acceptance retention ranges from 94.9% to 102.8%, while total KV savings range from 2.9% to 9.9%. Figure 3 shows that uniform K4V4 often saves more memory but can retain less than half of native acceptance; the sensitivity-aware solutions stay near the 100% line.

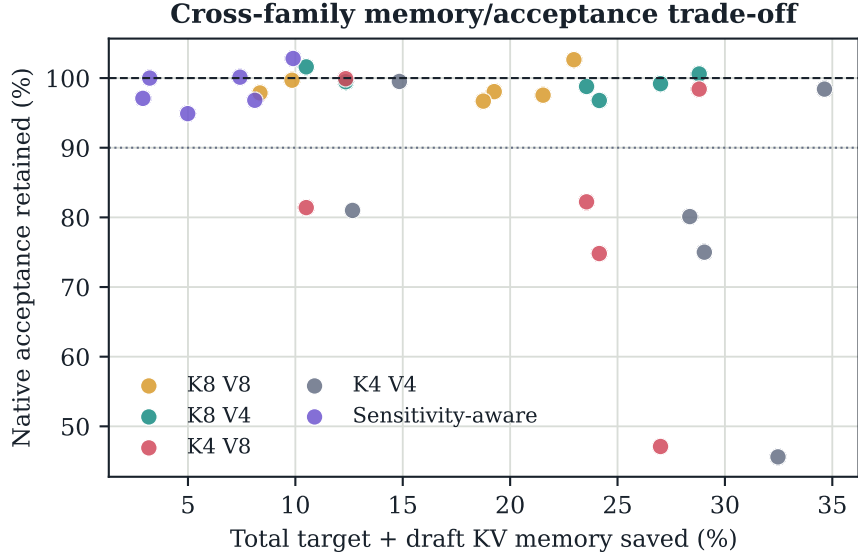


Figure 3: Memory–acceptance trade-off at 1K. Sensitivity-aware allocations are conservative; K8V4 provides a stronger uniform operating point than K4V8.

5.4 FINITE-PRECISION VERIFICATION AUDIT

The target cache and weights remain BF16 and are never quantized. Nevertheless, multi-token verification and tokenwise greedy decoding invoke different GPU kernel shapes. Across 2,010 prompt–configuration outputs, 308 sequence disagreements occur where either target path has top-1 margin at most 10^{-3} , and 33 (1.64%) remain above that threshold. Focused diagnostics found no causal-mask leakage and no cache-crop top-1 error; the remaining flips arise from small BF16 logit drift (e.g., KL below 10^{-2} in traced cases). We therefore do not claim bitwise equality to tokenwise BF16 greedy decoding, while the target verification rule itself remains unquantized.

5.5 MOTIVATING NEGATIVE RESULT: DIRECT CROSS-MODEL CACHE MAPS

Our earlier approach attempted to replace parts of the draft cache with linear maps from target-model KV states. Sharing the top four of 28 draft layers reduced total KV memory by only 6.25% and reduced acceptance from 0.497 to 0.455 in the cached-quality benchmark. Sharing every layer degraded acceptance much more. This failure motivated the current direction: preserve each model’s native cache semantics, compress the cheaper draft state, and optimize the quantity that matters for speculative decoding. Concurrent work on cross-model prefill reuse also finds that transfer quality depends strongly on error placement and often benefits from RoPE removal, cross-layer selection, or nonlinear maps (Heo et al., 2026).

6 RELATED WORK

KIVI establishes that key and value caches have different outlier structures and uses per-channel K and per-token V quantization (Liu et al., 2024). KV-AdaQuant argues for allocating more precision to keys based on norm and quantization sensitivity (Hariri et al., 2025). Our work does not claim that K/V asymmetry is new; it tests and optimizes that asymmetry under speculative acceptance, where a single changed proposal can truncate an entire draft block.

QuantSpec combines self-speculative decoding with a hierarchical 4-bit KV cache (Tiware et al., 2025). QSpec combines complementary model quantization schemes and reuses model state between drafting and verification (Zhao et al., 2025). In contrast, we study standard two-model speculative decoding and allocate draft-cache precision by measured acceptance sensitivity.

7 LIMITATIONS AND NEXT EXPERIMENTS

The present quantizer uses per-vector symmetric fake quantization. It does not yet implement packed storage, residual windows, per-channel key grouping, or fused attention kernels; therefore analytical cache bytes, not wall-clock latency, are the reliable systems metric. Although the campaign spans six pairs and includes two C4 seeds for three pairs, prompt counts remain modest, especially at 8K–16K. Broader PG19/LongBench evaluation and task-conditioned code and mathematics prompts are needed. Our BF16 audit also shows that mathematically distribution-preserving speculation need not be bitwise identical to tokenwise decoding across kernel shapes; higher-precision and production-kernel audits remain important.

The additive allocation in Equation 5 ignores interactions between compressed components. We plan to fit a small risk predictor over joint allocations and compare it against uniform precision, KIVI-style quantization, KV-AdaQuant, QuantSpec, and random allocations at matched memory. A packed-cache kernel in vLLM or SGLang is required before making end-to-end throughput and serving-capacity claims.

8 CONCLUSION

Draft KV compression should be evaluated by the speculative work it preserves, not reconstruction error alone. Across six model pairs, K8V4 wins 19 of 20 equal-memory comparisons, showing that keys are acceptance-critical routing state while values usually tolerate more aggressive compression. Acceptance-aware profiling preserves near-native draft quality with conservative savings, and the Qwen results show that substantially larger savings are possible. These results motivate packed asymmetric kernels and interaction-aware allocation optimized directly for speculative acceptance.

REFERENCES

- Charlie Chen, Sebastian Borgeaud, Geoffrey Irving, Jean-Baptiste Lespiau, Laurent Sifre, and John Jumper. Accelerating large language model decoding with speculative sampling. *arXiv preprint arXiv:2302.01318*, 2023.
- Mohsen Hariri, Lam Nguyen, Sixu Chen, Shaochen Zhong, Qifan Wang, Xia Hu, Xiaotian Han, and Vipin Chaudhary. More for keys, less for values: Adaptive KV cache quantization. *arXiv preprint arXiv:2502.15075*, 2025.
- Taekyung Heo, Rasoul Shafipour, Ritchie Zhao, Maximilian Golub, Mohammad Mahdi Kamani, Ritika Borkar, Makesh Tarun Chandran, Pantea Zardoshti, and Bitu Darvish Rouhani. Cross-model KV cache transfer in LLM families: A closed-form linear mapping for prefill reuse. *arXiv preprint arXiv:2608.03893*, 2026.
- Yaniv Leviathan, Matan Kalman, and Yossi Matias. Fast inference from transformers via speculative decoding. In *Proceedings of the 40th International Conference on Machine Learning*, pp. 19274–19286, 2023.
- Zirui Liu, Jiayi Yuan, Hongye Jin, Shaochen Zhong, Zhaozhuo Xu, Vladimir Braverman, Beidi Chen, and Xia Hu. KIVI: A tuning-free asymmetric 2bit quantization for KV cache. In *Proceedings of the 41st International Conference on Machine Learning*, pp. 32332–32344, 2024.
- Rishabh Tiwari, Haocheng Xi, Aditya Tomar, Coleman Hooper, Sehoon Kim, Maxwell Horton, Mahyar Najibi, Michael W. Mahoney, Kurt Keutzer, and Amir Gholami. QuantSpec: Self-speculative decoding with hierarchical quantized KV cache. In *Proceedings of the 42nd International Conference on Machine Learning*, pp. 59668–59686, 2025.
- Juntao Zhao, Wenhao Lu, Sheng Wang, Lingpeng Kong, and Chuan Wu. QSpec: Speculative decoding with complementary quantization schemes. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pp. 4779–4795, 2025.